



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: B. Tech

Semester: IV

Name of the Paper: Finite Automata and Formal Languages

Paper Code: TCS-402

Time: 1.5 hour

Maximum Marks: 50

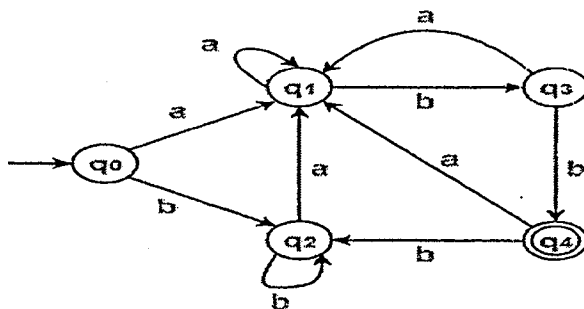
Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1.

(10 Marks)

- a. Minimize the following DFA using Myhill-Nerode theorem.



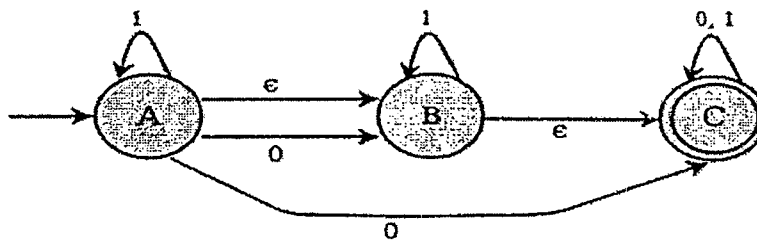
OR

- b. Design a NFA that accepts the strings having third last symbol from the last is 1, over input 0, 1. Also convert the NFA into equivalent DFA.

Q2.

(10 Marks)

- a. Convert the following Epsilon-NFA to DFA..



OR

- b. Obtain the grammar $L(G)$ from language $L = \{0^i 1^j, i \neq j, i, j > 0\}$.

Q3.

(10 Marks)

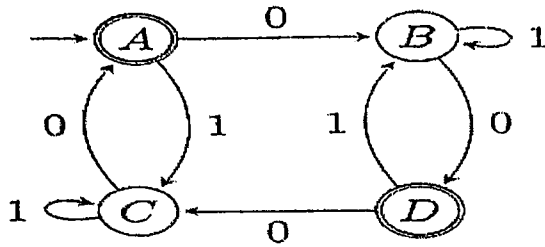
- a. Construct a moore machine that takes set of all strings over a, b as input and prints '1' as output for every occurrence of 'ab' as a substring. Also convert the Moore to equivalent Mealy machine.

OR

- b. Obtain the regular expression by using Arden's theorem for the below FA.



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Q4.

(10 Marks)

- a. Prove that regular grammars/languages are closed under intersection (by taking two finite automata's).

OR

- c. Prove that language $L = \{ww, w \in (0,1^*)\}$ is not regular using pumping lemma.

Q5.

(10 Marks)

- a. Construct the FA from regular expression $00^*(0^*+1)^*+1^*1(0+1^*)$.

OR

- b. Construct a DFA that accept strings of even number of a's and odd number of b's.